

Reinforcement Learning

Lab Assignment - 3

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Task 1: Formulating the GridWorld Environment

RL Component	Description
States	S1, S2, S3, S4, S5 and terminal goal state G
Actions	Up (↑), Down (↓), Left (←), Right (→)
Reward Structure	-1 for every normal movement; +10 when the goal state G is reached
Terminal State(s)	G is the terminal state
Goal of the Agent	Reach G from S1 while maximizing cumulative discounted reward and minimizing unnecessary steps

State Transition Structure

The important transitions are:

Current State	Action	Next State	Reward
S1	→	S2	-1
S1	↓	S4	-1
S2	→	S3	-1
S2	↓	S5	-1
S2	←	S1	-1
S3	↓	G	10
S3	←	S2	-1
S4	→	S5	-1
S4	↑	S1	-1
S5	→	G	10
S5	↑	S2	-1
G	Any	G	0

For Policy Evaluation:

State	Policy Action
S1	→
S2	→
S3	←

S4	→
S5	→
G	Terminal

Therefore, starting from S1, this policy does **not** reach the goal. This makes the difference between Policy Evaluation and Value Iteration clear.

Policy Evaluation

For a deterministic policy, Bellman expectation equation and let's put gamma as 0.9.

Value Function at Different Iterations:

Iteration	S1	S2	S3	S4	S5	G
1	-1.000	-1.000	-1.000	-1.000	10.000	0
2	-1.900	-1.900	-1.900	8.000	10.000	0
3	-2.710	-2.710	-2.710	8.000	10.000	0
4	-3.439	-3.439	-3.439	8.000	10.000	0
5	-4.095	-4.095	-4.095	8.000	10.000	0

The values of S1, S2 and S3 progressively become more negative because the evaluated policy continually moves between S2 and S3 without reaching the positive terminal reward.

Task 2: Policy Evaluation Observation Table

Using a convergence threshold of:

$\Delta < 0.001$

Iteration	Maximum Change (Δ)	Convergence Status
1	10.0000	Not converged
2	9.0000	Not converged
3	0.8100	Not converged
4	0.7290	Not converged
5	0.6561	Not converged

The maximum change decreases with successive iterations, indicating convergence toward a stable value function. With the selected threshold of 0.001, the policy evaluation converges at approximately **iteration 67**.

At convergence:

The negative value of S1–S3 indicates that repeatedly following the given policy is undesirable because it incurs continuous step penalties without reaching the goal.

State	Approximate Value
S1	-9.991
S2	-9.991
S3	-9.991
S4	8.000
S5	10.000
G	0

Task 3: Value Iteration

Value Iteration uses the Bellman Optimality Equation.

Unlike Policy Evaluation, Value Iteration does not evaluate one fixed policy. Instead, it repeatedly selects the action with the highest expected value.

Value Iteration Observations

Iteration	S1	S2	S3	S4	S5	Maximum Change
1	-1.0	-1.0	10.0	-1.0	10.0	10.0
2	-1.9	8.0	10.0	8.0	10.0	9.0
3	6.2	8.0	10.0	8.0	10.0	8.1
4	6.2	8.0	10.0	8.0	10.0	0.0

The optimal value function is therefore:

$$V^*(S1)=6.2, V^*(S2)=8.0, V^*(S3)=10, V^*(S4)=8.0, V^*(S1)=6.2, V^*(S5)=10, V^*(G)=0$$

Optimal Policy

The corresponding optimal policy is:

State	Optimal Action	State Value
S1	→	6.2
S2	→	8.0
S3	↓	10.0
S4	→	8.0
S5	→	10.0

Optimal Policy Visualisation

S1 ->	S2 →	S3 ↓
S4→	S5 →	G GOAL

Starting from S1, the optimal route is:

S1 → S2 → S3 → G

There is also an equally short route:

S1 → S4 → S5 → G

Both require three actions and produce a total reward of:

$$-1-1+10=8$$

For the table above, **Right is selected as the tie-breaking action at S1.**

Task 4: Optimal Path Analysis

For the random policy, a fixed random seed of **42** is used so that the result is reproducible. The maximum episode length is set to 50 steps.

Random Policy

One execution produced:

S1 → S1 → S1 → S1 → S4 → S4 → S4 → S1
→ S1 → S2 → S2 → S2 → S2 → S5 → S5
→ S2 → S5 → G

- Path length = 17 steps
- Total reward = -6
- Goal reached = Yes

The random agent makes several invalid or unnecessary movements before eventually reaching the goal.

Evaluated Policy

The evaluated policy produces:

S1 → S2 → S3 → S2 → S3 → S2 → S3 → ...
It never reaches G.

For a 50-step episode:

- Path length = 50
- Total reward = -50
- Goal reached = No

Optimal Policy

The optimal policy produces:

$S1 \rightarrow S2 \rightarrow S3 \rightarrow G$

- Path length = 3
- Total reward = 8
- Goal reached = Yes

Completed Observation Table

Policy Type	Path Length	Total Reward	Goal Reached
Random Policy	17	-6	Yes
Evaluated Policy	50*	-50	No
Optimal Policy	3	8	Yes

* The evaluated policy is actually non-terminating from S1; 50 is the imposed episode limit used for comparison.

Analysis Questions

1. What is a state in GridWorld?

A state represents the current position of the agent in the GridWorld. In this implementation, the states are S1, S2, S3, S4, S5 and the terminal goal state G.

2. What actions can the agent perform?

The agent can perform four directional actions:

- Up
- Down
- Left
- Right

An action that would take the agent outside the grid leaves it in the same state.

3. What is the purpose of Policy Evaluation?

Policy Evaluation calculates the **state-value function** $(V^{\pi}(s))$ for a fixed policy. It determines the expected cumulative discounted reward obtained when the agent follows that particular policy.

In this experiment, Policy Evaluation shows that the selected policy is poor because it creates a loop between S2 and S3 and fails to reach the goal from S1.

4. How does Value Iteration differ from Policy Evaluation?

Policy Evaluation evaluates an already specified policy.

Value Iteration directly searches for the optimal value of each state by considering all available actions and selecting the action with the maximum expected value.

Therefore:

Policy Evaluation



Given Policy



Calculate $V\pi(s)$

whereas:

Value Iteration



Consider all actions



Calculate $V^*(s)$



Select best action



Optimal Policy

5. What is the Bellman Optimality Equation?

The Bellman Optimality Equation is:

$$V^*(s) = \max_a \left[R(s, a) + \gamma \sum_{s'} P(s'|s, a) V^*(s') \right]$$

For this deterministic

GridWorld:

$$V^*(s) = \max_a [R(s, a) + \gamma V^*(s')]$$

It finds the maximum achievable expected return from each state.

6. Which policy reached the goal using the fewest steps?

The **Optimal Policy** reached the goal using the fewest steps.

The optimal path is:

6. Which policy reached the goal using the fewest steps?

The **Optimal Policy** reached the goal using the fewest steps.

The optimal path is:

$S1 \rightarrow S2 \rightarrow S3 \rightarrow G$
with **3 steps**.

An equally short alternative is:

$S1 \rightarrow S4 \rightarrow S5 \rightarrow G$

7. Why is the optimal policy superior to a random policy?

The optimal policy considers the estimated future rewards of all available actions and selects the action leading toward the highest-value state. Therefore, it avoids unnecessary movements and reaches the goal in the minimum number of steps.

The random policy has no knowledge of the state values and can move away from the goal or repeatedly attempt invalid movements.

In this experiment:

Policy	Steps	Reward	Goal
Random	17	-6	Yes
Evaluated	50*	-50	No
Optimal	3	8	Yes

Thus, the optimal policy is substantially more efficient.

8. How would obstacles affect the optimal path?

Obstacles would remove certain transitions from the state space. The agent could no longer move through blocked cells, so Value Iteration would need to find an alternative route.

For example:

$S1 \rightarrow S2 \rightarrow X$
obstacle

would prevent the agent from continuing through that route. The optimal policy would instead select another available direction, potentially producing a longer path.

Obstacles can therefore:

- Increase the minimum number of steps.
- Change the optimal action at several states.
- Reduce the state values of states farther from the goal.
- Force the agent to take alternative routes.
- Potentially make some states unreachable from the goal.

The supplied Neuromatch GridWorld implementation explicitly models walls by making an attempted movement into a wall result in the agent remaining in its current cell.

Key Convergence Observation

The Policy Evaluation results show:

Iteration 1 $\Delta = 10.0000$

Iteration 2 $\Delta = 9.0000$ Iteration 3 $\Delta = 0.8100$ Iteration 4 $\Delta = 0.7290$ Iteration 5 $\Delta = 0.6561$

The maximum change decreases as iterations continue. Therefore, the value function is moving toward a fixed point.

However, after only five iterations it has **not yet reached the selected convergence threshold of 0.001**. It converges at approximately iteration 67.

Value Iteration converges much faster for this small deterministic environment, reaching the stable values by iteration 4.

OUTPUT:

[illegible]